



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 11

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Program:

```
def heapify(arr, n, i):  
    largest = i  
    left = 2 * i + 1  
    right = 2 * i + 2  
    if left < n and arr[left] > arr[largest]:  
        largest = left  
    if right < n and arr[right] > arr[largest]:  
        largest = right  
    if largest != i:  
        arr[i], arr[largest] = arr[largest], arr[i]  
        heapify(arr, n, largest)  
  
def heap_sort(arr):  
    n = len(arr)  
    for i in range(n // 2 - 1, -1, -1):  
        heapify(arr, n, i)  
    for i in range(n - 1, 0, -1):  
        arr[0], arr[i] = arr[i], arr[0]  
        heapify(arr, i, 0)  
    return arr
```

```
arr = [4, 10, 3, 5, 1]
```

```
print("Sorted array:", heap_sort(arr))
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def heapify(arr, n, i): 2 largest = i 3 left = 2 * i + 1 4 right = 2 * i + 2 5 6 if left < n and arr[left] > arr[largest]: 7 largest = left 8 9 if right < n and arr[right] > arr[largest]: 10 largest = right 11 12 if largest != i: 13 arr[i], arr[largest] = arr[largest], arr[i] 14 heapify(arr, n, largest) 15 16 17 def heap_sort(arr): 18 n = len(arr) 19 20 for i in range(n // 2 - 1, -1, -1): 21 heapify(arr, n, i) 22 23 for i in range(n - 1, 0, -1): 24 arr[0], arr[i] = arr[i], arr[0] 25 heapify(arr, i, 0) 26 27 return arr</pre>			Sorted array: [1, 3, 4, 5, 10]